Flavor Explorers

Does pleasure come at the cost of health?

Data Visualization Course Process Book

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Introduction

A common assumption is that living a good and healthy life often requires sacrificing pleasure and enjoyment. Whether in lifestyle choices, habits, or consumption, people frequently perceive a trade-off between what is beneficial and what is enjoyable. In our work, we examine this question through the lens of food, a central and deeply cultural element of human life.

In particular, we investigate the effects of food on both health and mood, emphasizing the relationship between nutritional value and the pleasure and emotional satisfaction food provides. Alongside our other methods, we model foods based on shared flavor compounds between ingredients. By identifying ingredients with similar flavor profiles, we can derive healthier alternatives to traditionally unhealthy components while preserving the sensory experience and enjoyment associated with the original dishes. Through this approach, we aim to explore whether healthier eating can be achieved without sacrificing taste, mood enhancement, and overall culinary satisfaction.

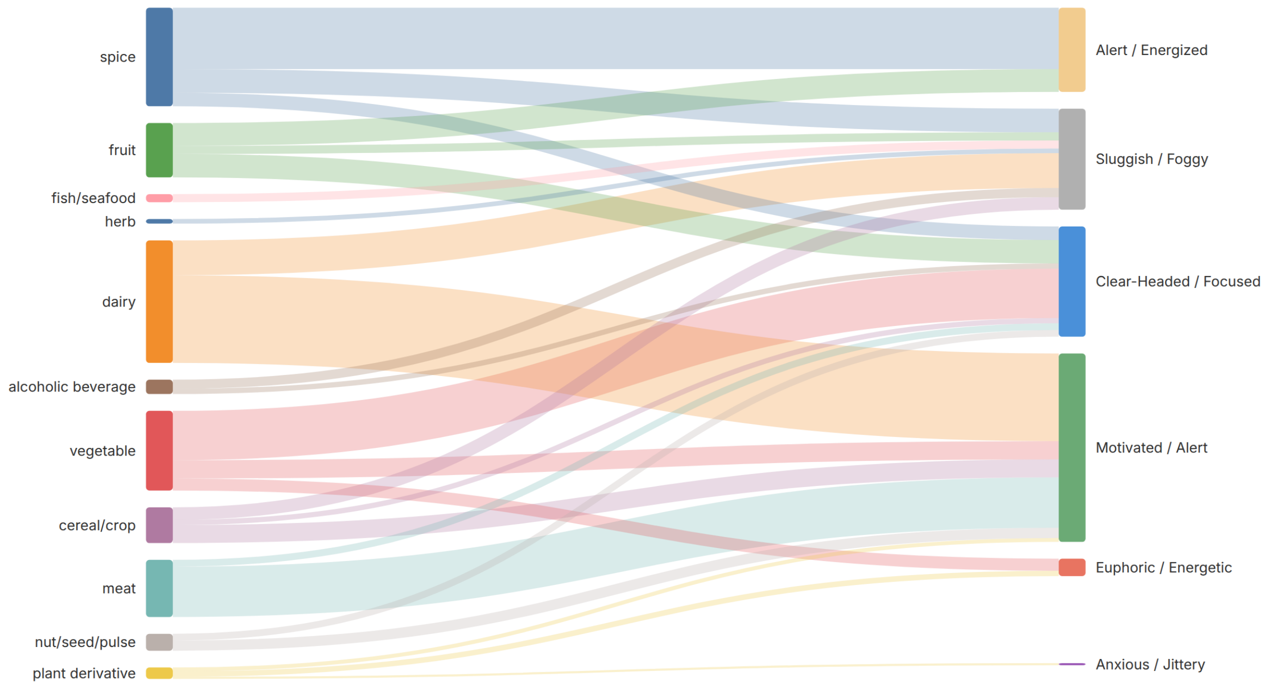

Figure 1: The flavor compounds of ingredients lead to different emotions and moods.

Journey

Our starting point

Obtaining suitable data for this study was a significant challenge, as no existing dataset directly captured the relationships between food, health, mood, and flavor in the form required for our analysis. Furthermore, both health and mood are inherently subjective concepts, and there is no universally accepted or normalized metric that quantifies them in a consistent way.

To address these challenges, we combined multiple standalone datasets into a unified framework connected through shared flavor compounds and ingredient relationships. One of our contributions is the meaningful integration and enrichment of these heterogeneous datasets into a more coherent and usable representation of **food relationships**.

Our work is primarily based on the Kaggle Recipes and the FlavorDB datasets. We first processed and standardized the Kaggle Recipes dataset in order to match its ingredients against FlavorDB's 1,530 canonical ingredients. This resulted in a dataset containing 25,009 **unique recipes** constructed from 9,840 **unique ingredients**.

We then enriched this dataset with both health and mood information. Health scores were generated by using a large language model (LLM) to assign each ingredient a score in the range of $[-1, 1]$, representing its estimated nutritional impact. Mood scores, in contrast, were derived from chemistry-based evidence: for each ingredient, we summed the weights of mood-active compounds identified through the FlavorDB compound relationships. To ensure comparability and consistency across ingredients, the resulting mood scores were also normalized to $[-1, 1]$. Through this process, we constructed a **dataset** that jointly captures **nutritional**, **emotional**, and **flavor-based** dimensions of food.

Bringing Our Vision to Life

We started our design process with extensive **rapid prototyping** using pen and paper. Each team member worked individually to fully express their own ideas without being influenced by others. Our first comparison of the sketches revealed different expectations and, in some cases, contrasting ideas for the visualizations. However, one common element appeared across nearly all sketches: the decision to adopt a **scrollytelling** approach.

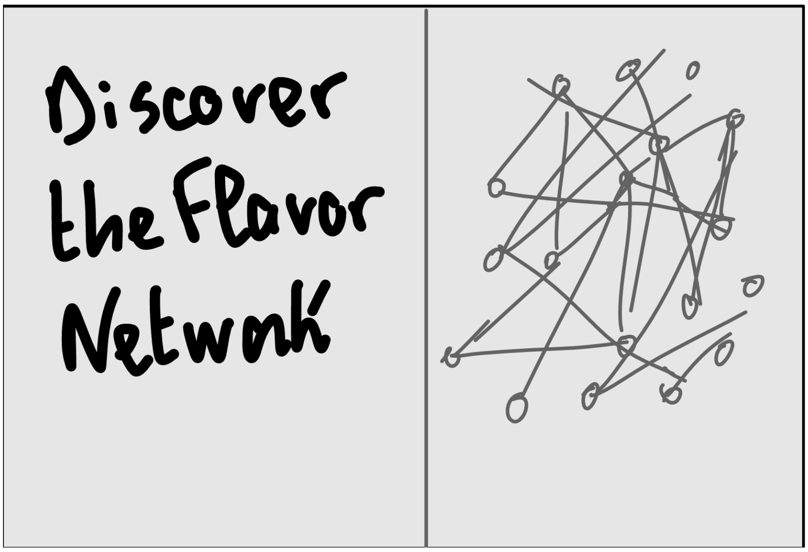

Figure 2: The very first sketch of the molecular network.

After agreeing on several aspects of our different design proposals, we created a comprehensive sketch of the entire website. To further materialize our ideas and consolidate our shared vision for the ongoing design process, we developed another **prototype** using the interface design platform **Figma**. Through this prototype, we resolved the remaining design disagreements and refined our vision of the final product.

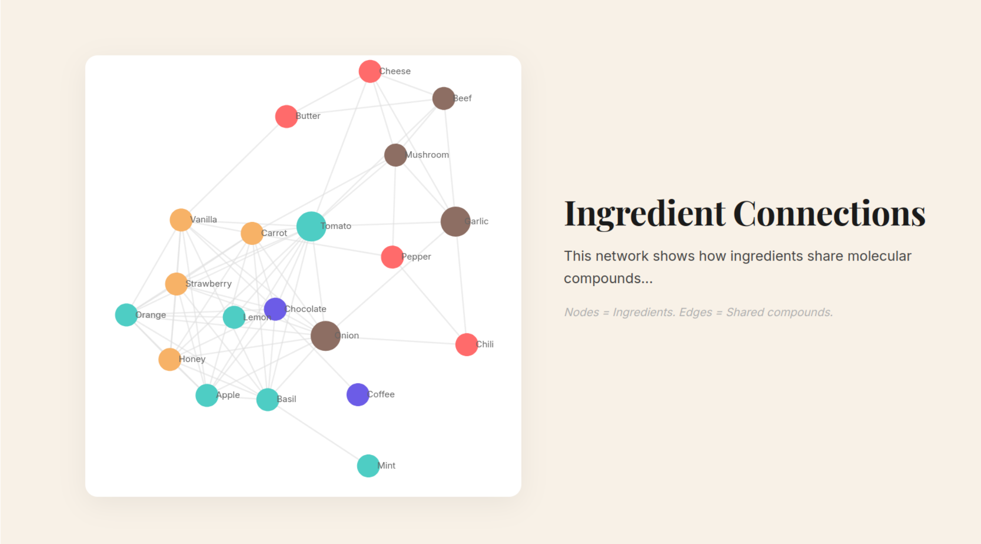

Figure 3: The molecular network in the Figma prototype.

We decided on the following figures and interactive game to help users gain intuitive and visual insights into the relationships discovered in this study, highlighting connections between ingredients through **shared flavor compounds** and positioning foods within the broader **health–mood space** identified in our analysis.

The Story Behind the Numbers A scatter plot visualizing food categories within the health–mood space, showing where recipe categories are positioned and using bubble size to represent their relative volume. This provides users with an intuitive first impression of how different categories are distributed across the health–mood landscape.

Where do ingredients actually sit? A heatmap illustrating the distribution of ingredients across the health–mood space, complementing the initial impressions and insights gained from the preceding food category overview. The visualization also reveals that most ingredients achieve relatively high health and mood scores.

Who are the double champions? Two ranked ingredient lists, one sorted by health score and the other by mood score, highlighting ingredients that appear in both rankings. This enables users to identify ingredients associated with simultaneously high health and mood scores.

Does it hold recipe by recipe? A scatter plot of recipes from selected recipe categories, allowing users to explore how recipes within each category are distributed across the health–mood space. The visualization also shows that recipes within the same category tend to form closer clusters in the health–mood space.

Which cuisines play it safe? A scatter plot-based recipe-by-category visualization of cuisine-specific categories, where circle size is dependent on the number of recipes in each category, allowing users to estimate where different cuisines are positioned.

The full category landscape A scatter plot of all 140 recipe categories positioned in the health–mood space, where the circle size is independent of the number of recipes in each category.

Healthy Ingredients, Better Moods? A violin plot showing the distribution of ingredients across four categories, highlighting that healthier ingredients in expectation are also associated with a higher mood score.

From ingredient to feeling A Sankey diagram linking ingredient categories to emotions and feelings, illustrating what emotional outcomes users might can expect from each category of ingredients.

Connected at the Molecular Level A network graph linking ingredients that share flavor compounds, with node colors indicating health levels, allowing users to identify ingredients with similar flavor profiles and potential substitutions that preserve comparable taste.

Build a recipe The final element is an interactive game where users can explore how mood and health scores change when combining different ingredients. Users can adjust the quantity of each ingredient, with each contribution weighted according to its selected share, dynamically updating the overall scores.

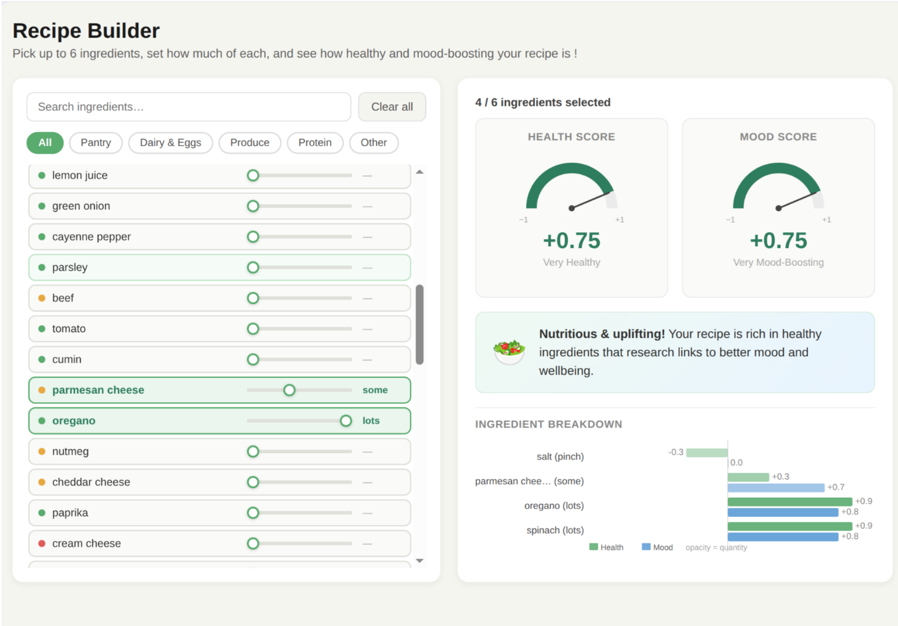

Figure 4: Ingredient game illustrating how the overall health score is built from individual ingredients, weighted according to the quantity added to the recipe.

Making It All Happen

We designed our website as a space for exploration and discovery, reflecting the inherently **subjective nature of food**, where different users have different needs, preferences, and motivations. Rather than presenting fixed conclusions, we aim to enable users to **derive** their **own insights** from the data and form their own interpretations of the results.
Our scrollytelling approach gradually guides users through the dataset and key findings in a structured, step-by-step flow. This format encourages engagement by **revealing information progressively**, while preserving a strong sense of orientation and control. Users can easily navigate between sections and return to earlier content at any point. The familiarity of the scrollytelling pattern also helps make the experience intuitive, accessible, and easy to follow.

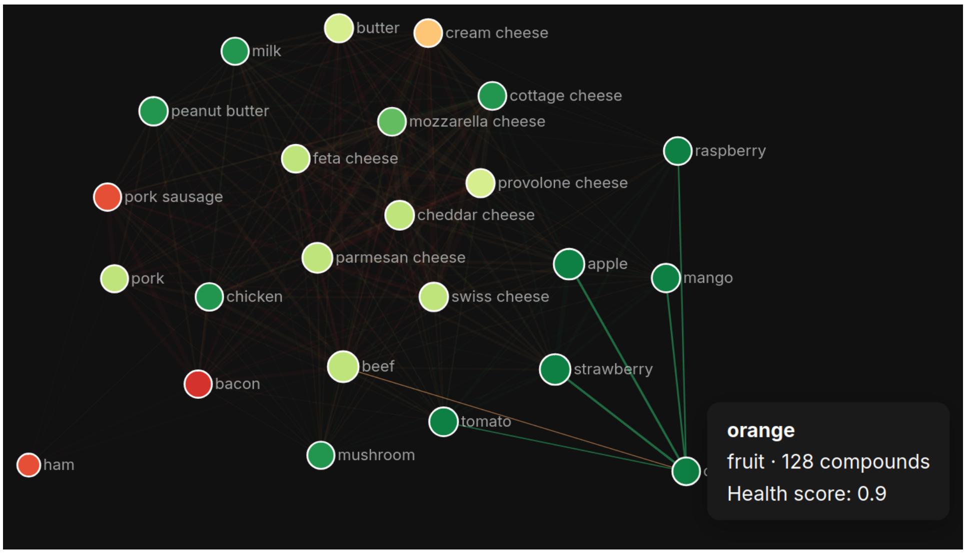

Figure 5: The molecular network on the final website.

The visualizations are implemented using **D3.js**, following the principles and techniques learned throughout the course. This includes applying best practices for data binding, scales, axes, and interactive elements to ensure clarity and responsiveness in the presentation of the data.
Most of the data processing is also handled directly in D3.js, allowing for a streamlined workflow where data transformation, aggregation, and visualization logic are closely integrated. This approach helps maintain consistency between the raw dataset and its visual representation.
The game was developed as a standalone website, built entirely using HTML, CSS, and JavaScript. It was fully integrated as a self-contained web application, which ensured a narrow and focused tech stack with no external frameworks or dependencies. This approach kept the implementation lightweight, transparent, and easy to maintain.
For the scrollytelling experience, JavaScript is used to manage the narrative flow and user interactions. It controls section transitions, scroll-based triggers, and the coordination between text and visual updates, ensuring a smooth and engaging step-by-step storytelling experience.

Our final product is a scrollytelling website that allow users to learn about, interactively explore, and critically examine their food habits.

Challenges

Food diversity involves bringing together a wide range of recipes, which can sometimes lead to comparisons that are not necessarily meaningful for users, e.g., recipes containing baby milk may not be relevant to most people's needs. On the other hand, this diversity also creates opportunities to learn new things and gain insights into unfamiliar recipes and ingredients from different cuisines that users may not have encountered before.

Since a single suitable data source was not available, we had to combine multiple datasets to build a coherent and usable foundation for our project. This integration process introduced several difficulties, such as inconsistencies in the data structure and non-normalized molecule names, which made direct matching and aggregation difficult.
Because no clean, fully normalized dataset existed, we had to make the decision to further process and enrich the data ourselves. This included manually defining labeling rules and expanding the dataset using an LLM to help standardize and structure the information more consistently. This approach allowed us to create and contribute a more unified dataset suitable for analysis and visualization, despite the limitations of the original sources.

We had no real prior experience in web development, so much of the work involved learning new concepts from scratch. This resulted in a steep learning curve, particularly when dealing with programming challenges such as common JavaScript pitfalls. Built-in type coercion was a common source of errors, while CSS layout and positioning issues also proved to be frequent challenges.

Peer Assessment

All team members contributed extensively to nearly every phase of the project, establishing a deeply collaborative environment. Project scoping, dataset parsing, and core architectural choices were debated and finalized during intensive on-campus and remote synchronization sessions. To optimize our development speed and address immediate blockers, we adopted a fluid "work-stealing" strategy, enabling team members to seamlessly pick up tasks, debug teammate code, and ensure balanced workloads throughout the semester.
To ensure accountability and transparent grading, the specific ownership and individual contributions are detailed below:

Imane Benkamoun

Imane served as the structural backbone and creative strategist for the project. She conceptualized the overarching thematic framework, steering the investigation away from basic nutritional tracking toward the complex "pleasure vs. health" cultural paradox. Her core contributions include:

- **Creative Direction & Storytelling Architecture:** Formulated the foundational user journey and designed the scrollytelling flow, mapping technical visual assets to an intuitive narrative thread that unrolls progressively as the user scrolls.
- **Core Infrastructure Development:** Managed the structural Git repository setup and built the central web application framework, guaranteeing a cohesive platform for embedding standalone visual assets.
- **Visual Components:** Spearheaded the design and front-end execution of the multi-view scatter plots, managing data scales and responsive grid alignments across multi-dimensional visual landscapes.

Quirin Bitter

Quirin acted as the primary visual architect and core plot engineer, translating abstract concepts into high-fidelity components. His core contributions include:

- **Advanced Visualization Pipelines:** Programmed the complex D3.js visualization charts, implementing the intricate graph physics layout for the molecular flavor network and designing the ingredient-to-feeling Sankey diagrams.
- **Fine-Grained UI/UX Design:** Translated hand-drawn concepts into precise Figma digital prototypes, establishing the project's visual identity, typographic hierarchy, and high-contrast styling choices.
- **Interactivity Engineering:** Programmed tailored transition triggers, dynamic hover highlights, and coordinate tooltips within the network and ranked components, ensuring low-latency interactive feedback.

Sofia Taouhid

Sofia served as the data engineer and logic specialist, anchoring the project's analytical integrity. Her core contributions include:

- **Data Pipelines & Heterogeneous Fusion:** Handled the extraction, transformation, and loading (ETL) steps required to link the Kaggle Recipes and FlavorDB corpuses, engineering custom standardization scripts to bridge naming gaps, and drove key data selection decisions through iterative visual exploration.
- **LLM Prompt Engineering & Compound Evaluation:** Authored the LLM pipeline for parsing and normalizing health scores, and structured the chemical-weight formulas to compute normalized mood indicators.
- **Visualization & Interactive Simulation Programming:** Designed and implemented the violin plot surfacing the correlation between nutritional quality and mood outcomes, and architected the vanilla JavaScript engine powering the "Build a Recipe" interactive game with real-time health and mood score updates.